

# Final Analytics & Decision Support Report

---

### Week 8 | Data Analytics Track

Final Integration, Validation & Project Showcase

## Executive Summary

This report presents the final Data Analytics contribution to the HealthConnect Clinic project, building on the analysis, dashboard development, testing and refinement completed in Weeks 5–7. The final output combines an interactive Power BI dashboard, validated KPIs, segment-level findings, business recommendations and documented cross-track integration.

Across 5,000 synthetic appointment records, the overall no-show rate was 48.5% and the lost slot rate was 53.7%. No-show rates increased across the booking lead-time bands, from 24.8% for appointments booked 0–3 days in advance to 60.5% for appointments booked more than 30 days in advance. Within the 30+ day group, the observed no-show rate was 67.9% for appointments with previous no-show history and 55.2% for those without it. These are descriptive findings and do not establish causation.

## Project Purpose and Business Question

The project addresses missed appointments and the appointment capacity lost when scheduled visits do not result in completed attendance. The guiding question is:

***How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?***

## Week 7 to Week 8 Transition

Week 7 focused on testing, refinement and end-to-end validation. The main output was a refined Power BI dashboard with interactive slicers and validated KPI behaviour. Testing covered the baseline state, booking lead time, previous no-show history, reminder channel, appointment type and distance. Two issues were identified: the Repeat No-Show KPI did not initially respond correctly to the “No prior no-show” selection, and the distance category visual did not follow the intended logical order. The DAX measure and category sort order were refined and retested. Week 8 focuses on final presentation, decision support and documenting the remaining integration dependency.

## Final Data Analytics Component and Readiness

The final component is the HealthConnect Advanced No-Show Analysis dashboard, supported by this report. It addresses the need to understand how appointment booking characteristics and patient history are associated with no-shows. The dashboard provides KPI cards and interactive slicers for booking lead time, previous no-show history, reminder channel and appointment type; distance is included in the analytical visuals.

The component is ready to be presented as a validated descriptive analytics and decision-support output. It is not a clinical decision system and does not independently prescribe interventions or establish causation.

## Dataset and Analytical Approach

The analysis used 5,000 HealthConnect appointment records. Relevant fields include booking lead days, appointment outcome, previous no-shows, reminder channel, reminder status, appointment type and distance to clinic. Power Query was used for data preparation and calculated fields; Power BI and DAX were used to develop KPI measures and interactive visuals. Figures were checked against the raw appointment data.

The booking lead-time bands were 0–3 days, 4–14 days, 15–30 days and 30+ days (more than 30 days). Segment results are descriptive rates calculated within each segment.

## Final KPIs

| KPI                  | Final value | Definition / interpretation                                          |
|----------------------|-------------|----------------------------------------------------------------------|
| Overall No-Show Rate | 48.5%       | No-show appointments divided by all appointments.                    |
| 30+ Day No-Show Rate | 60.5%       | No-show rate among appointments booked more than 30 days in advance. |
| Repeat No-Show Rate  | 55.4%       | No-show rate among appointments with previous no-show history.       |
| Lost Slot Rate       | 53.7%       | No-shows plus cancellations, divided by all appointments.            |

## Validated Analytical Findings

### Booking Lead Time

The no-show rate increased across the booking lead-time bands: 24.8% for 0–3 days, 32.6% for 4–14 days, 43.2% for 15–30 days and 60.5% for 30+ days. This pattern supports monitoring longer-lead-time appointments. It should be described as an association, not proof that lead time causes a no-show.

### Booking Lead Time and Previous No-Show History

Within the 30+ day segment, appointments with previous no-show history had a 67.9% no-show rate (1,015 appointments), compared with 55.2% for appointments without previous no-show history (1,410 appointments). The difference is 12.7 percentage points. Both groups were above the overall 48.5% rate. This combined segment is a useful candidate for closer monitoring and further predictive-model evaluation.

### Reminder Channel

For appointments booked 30+ days in advance, observed no-show rates were 61.9% for Email, 63.6% for None, 58.0% for SMS and 60.2% for WhatsApp. These descriptive differences can support ongoing monitoring by segment; they do not demonstrate that a particular channel causes better or worse attendance.

### Appointment Type

Within the 30+ day segment, Follow-up appointments had the highest observed no-show rate at 65.1%. Diagnostic Test appointments were 59.6%, Specialist Consultation 59.2% and General Consultation 58.1%. These differences may guide further review of appointment-specific support needs, without implying causation.

### Distance to Clinic

Within the 30+ day segment, observed no-show rates were 66.4% for distances over 15 km, 61.4% for under 5 km, 58.2% for 5–15 km and 64.3% for unknown distance. The pattern is not monotonic, so distance should be treated as an additional segmentation variable rather than a simple linear explanation. Missing distance values also limit interpretation.

## Testing, Validation and Refinement

Week 7 testing covered the baseline dashboard state, lead-time selections, previous no-show history, reminder channel, appointment type and distance. Core KPI calculations and segment-level figures were checked against the raw data. The documented tests passed after the identified refinements.

| Issue / test area                   | Refinement                                                           | Retest outcome                                                |
|-------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------|
| Repeat No-Show KPI slicer behaviour | Updated DAX logic to respect the Previous No-Show History selection. | No prior no-show selection returns 0.0%; passed after retest. |
| Distance category sort order        | Applied a custom sort column in Power Query.                         | Distance categories display in the intended logical order.    |
| Core KPIs and segment analysis      | Compared dashboard values with raw-data calculations.                | Validated for baseline and tested segment selections.         |

## Business Insights and Decision Support

- Longer booking lead time is associated with a higher observed no-show rate in the dataset.
- Previous no-show history provides additional segmentation information within the 30+ day booking group.
- Reminder channel, appointment type and distance show differences across segments and can inform monitoring and further investigation.
- The interactive dashboard enables users to explore these patterns rather than relying only on static summary figures.

## Recommendations

**Monitor long-lead-time bookings:** Review appointments booked more than 30 days in advance as a distinct monitoring segment.

**Combine lead time and history for review:** Use previous no-show history alongside booking lead time to identify segments for additional patient-support review.

**Explore appointment-specific support:** Review whether follow-up appointments have particular scheduling or communication needs, using further evidence before changing policy.

**Evaluate reminder approaches by segment:** Continue monitoring reminder-channel outcomes by segment; do not assume observed differences are caused by the channel.

**Evaluate interventions using future outcomes:** If HealthConnect pilots changes, compare subsequent results against a suitable baseline and document the evaluation method.

## Cross-Track Integration

The Data Analytics finding on the combined 30+ day booking and previous no-show segment was shared with the Data Science track. The collaborator identified a possible interaction feature combining long booking lead time and previous no-show history as a candidate for testing in the predictive model. This documents a handoff from descriptive analytics to model development.

The available evidence does not include a subsequent model-performance result or confirmation that the candidate feature improved predictive performance. This remains an open integration item for the wider HC-POD walkthrough. The analytics output can also support Project Management by providing validated findings and clearly stated limitations for final business recommendations.

## Final Integration and Presentation

The Data Analytics component contributes validated appointment analysis, KPI definitions, an interactive dashboard, segment-level findings, recommendations and limitations. In the end-to-end walkthrough, it should be presented as the evidence layer that helps explain the appointment problem and informs evaluation of the wider predictive, workflow and patient-support components.

For the final presentation, demonstrate the dashboard using the interactive slicers, particularly the 30+ day booking segment and the previous no-show history comparison. Distinguish completed analytics validation from open cross-track dependencies.

## Limitations

- The dataset is synthetic and intended for learning and analytical practice, not direct clinical decision-making.
- The analysis is descriptive; observed associations do not establish causation.
- Some distance and waiting-time values are missing or may be unreliable for appointments that were not attended.
- Reminder-channel comparisons may reflect differences between appointment groups and should not be interpreted as causal effects.
- No new verified Data Science model-performance result is included in the available final analytics evidence.

## Conclusion

The final Data Analytics work provides HealthConnect with a tested, interactive view of appointment no-show patterns. The clearest observed pattern is the increase in no-show rates across longer booking lead times, with an additional difference by previous no-show history within the 30+ day group. The dashboard and recommendations support focused monitoring and further evaluation while preserving the distinction between descriptive evidence and proven intervention effects.

